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Master thesis

Stress Level Estimation using Radar Sensor, EDA, and HRV measurements with Sensor Fusion and Machine Learning

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Abstract

Putting the abstract later, because I have to get into more details of the project.

Acknowledgments

This master thesis is done in collaboration with the Biomedical group at the Physics department of University of Oslo. I would like to thank my supervisors Ørjan Martinsen and Jie Hou for their patience, help and for making this project possible. I hope that this work will contribute to future innovations in stress- monitoring and managing field.

AI Disclaimer

This thesis was written with the assistance of AI-based tools (e.g., ChatGPT) for language refinement, structuring of text, and clarification of scientific writing. All scientific content, methodology, analysis, and conclusions are the author's own work. AI tools were not used to generate data, perform analysis, or produce scientific results.

Chapter 1

INTRODUCTION

1.1 Background

Stress is a major factor affecting human health, productivity, and overall quality of life, and is closely associated with a wide range of physical and psychological disorders. Prolonged or poorly managed stress has been linked to cardiovascular disease, depression, impaired cognitive performance, and reduced workplace efficiency. Consequently, reliable and continuous stress monitoring has become an important objective in modern healthcare, occupational safety, and personal well-being applications [1].

Traditional stress assessment relies primarily on self-report questionnaires and wearable physiological sensors, which can be subjective, intrusive, and inconvenient for long-term monitoring. In recent years, contactless sensing technologies have emerged as a promising alternative, offering the potential for unobtrusive and continuous physiological monitoring [2, 3].

Ultra-wideband (UWB) radar sensors, such as the XeThru X4M200, were originally developed for respiration and motion tracking but have demonstrated the capability to capture subtle physiological micro-movements related to cardiac and respiratory activity. These properties make radar-based sensing a compelling modality for non-contact stress monitoring[4].

1.2 Problem statement

Despite significant advances in radar-based physiological sensing and multimodal stress detection, existing approaches primarily rely either on radar-derived signals alone or on wearable sensor modalities in isolation. There is limited research that systematically integrates contactless radar-derived respiratory and motion features with wearable autonomic nervous system biomarkers, such as heart rate variability (HRV) and electrodermal activity (EDA), using machine learning–based sensor fusion strategies. Furthermore, most prior studies evaluate stress classification under constrained laboratory conditions, without fully investigating the complementary benefits of combining remote and contact-based sensing modalities[5, 6].

1.3 Research objectives

This thesis aims to develop a multimodal stress estimation framework that combines contactless UWB radar–derived respiratory and motion features with wearable

autonomic biomarkers, specifically heart rate variability (HRV) and electrodermal activity (EDA). The objective is to employ sensor fusion strategies and machine learning techniques to enable accurate, robust, and non-invasive stress level estimation under both mental and physical stress conditions[7, 8, 9].

1.4 Research questions

This thesis seeks to answer the following research questions:

- Can contactless radar-derived respiratory and motion features, when fused with wearable HRV and EDA biomarkers, improve stress level estimation compared to single-modality approaches?
- How do feature-level and decision-level sensor fusion techniques compare in stress classification performance?
- Can multimodal fusion improve robustness and generalizability compared to single-modality approaches?
- How do mental and physical stressors affect the performance of the proposed stress estimation framework?

1.5 Scope of the study

This study focuses on stress estimation using UWB radar, HRV, and EDA signals collected under controlled stress-inducing protocols. Radar is used for contactless extraction of respiration- and motion-related features, while HRV and EDA are obtained using wearable sensors. The work is limited to healthy adult participants and does not address clinical diagnosis of stress-related disorders. The research emphasizes methodological development, feature extraction, sensor fusion, and machine learning evaluation rather than medical treatment or long-term monitoring in uncontrolled environments.

1.6 Research contributions

This thesis contributes to the field of stress monitoring by proposing a novel multimodal framework that integrates contactless radar sensing with validated autonomic biomarkers. The main contributions include:

- Development of a synchronized multimodal data acquisition setup combining contactless radar sensing with wearable HRV and EDA measurements.
- Design of feature-level and decision-level sensor fusion strategies for stress estimation.
- Implementation and evaluation of machine learning models for multimodal stress classification.
- Comparative analysis of stress estimation under mental and physical stress conditions.

1.7 Thesis outline

This thesis is organized as follows. Chapter 2 presents the physiological background of stress and reviews related work in wearable and radar-based stress monitoring. Chapter 3 describes the experimental design, sensor setup, and stress induction protocol. Chapter 4 details the feature extraction, sensor fusion strategies, and applied machine learning models. Chapter 5 presents the experimental results and evaluation metrics. Chapter 6 discusses the findings, limitations, and future research directions.

Chapter 2

LITERATURE REVIEW

2.1 Definition of stress

Stress is the response of the body to a psychological or physical stressor, which can be a real or perceived threat that disrupts homeostasis. It is often related to fear and anxiety and thus serves as an alarm to an immediate danger or possible threat. In a more primitive sense, stress was necessary for our survival in ancient times, where predators, venomous animals, and natural disasters contributed to a life-threatening environment. In contemporary society, stress is no longer limited to life-threatening situations but is frequently associated with daily occupational, social, and cognitive demands. [10].

2.2 Physiological mechanism of stress

When encountering a stressor, the body and the brain initiate a complex, stress response designed to help the organism cope with the presented challenge. The stressor is detected and evaluated by brain regions such as the amygdala, hippocampus, and prefrontal cortex[11].
Amygdala is mainly responsible for fear. It processes things we see and hear, learns what is dangerous and uses it later to cause the feel of fear or similar emotions [12].
Hippocampus is responsible for learning and memory. It helps converting short-term memory to a long-term memory and contributes to environment awareness. Hippocampus and amygdala create together emotional response which is essential in stress response, by connecting emotions and memory together [13].
Prefrontal cortex plays a central role in managing your thoughts, emotions, and behaviors. It enables executive functions such as attention, decision-making, self-control, planning, and working memory [14].

2.3 Hypothalamus

Hypothalamus acts as the main control center of the stress response. After the amygdala and hippocampus process and evaluate the stressor, the hypothalamus receives this information and decides how the body should react. It then coordinates the next steps by sending signals that activate two major systems responsible for managing stress. These are the sympathetic-adrenomedullary (SAM) system, which triggers an immediate “fight-or-flight” reaction, and the hypothalamic–pituitary–adrenal (HPA) axis, which helps maintain the response over a longer period. Through these systems, the hypothalamus

ensures that the body can respond effectively to the stressor and later return to a balanced state once the threat has passed.

SAM and HPA

The SAM system is the body's first and fastest response to stress. When a stressor is detected, the hypothalamus immediately activates the sympathetic nervous system, which stimulates the adrenal medulla to release adrenaline and noradrenaline into the bloodstream. These hormones cause rapid physiological changes such as increased heart rate, elevated blood pressure, and faster breathing. The body becomes more alert and ready for action, preparing for a "fight-or-flight" reaction. This quick response allows the organism to deal with the immediate challenge, but it is meant to last shortly and to return to normal once the threat has passed.
The HPA axis on the other hand, represents the slower and longer-lasting part of the stress response. After the initial SAM activation, the hypothalamus releases corticotropin-releasing hormone (CRH), which prompts the pituitary gland to secrete adrenocorticotropic hormone (ACTH). ACTH travels through the bloodstream to the adrenal cortex, where it triggers the production and release of cortisol. Cortisol helps the body maintain energy balance, regulate immune activity, and manage prolonged stress situations. Once the stressor is gone, cortisol levels decrease, and feedback signals from the brain help restore homeostasis, bringing the body back to a stable state.

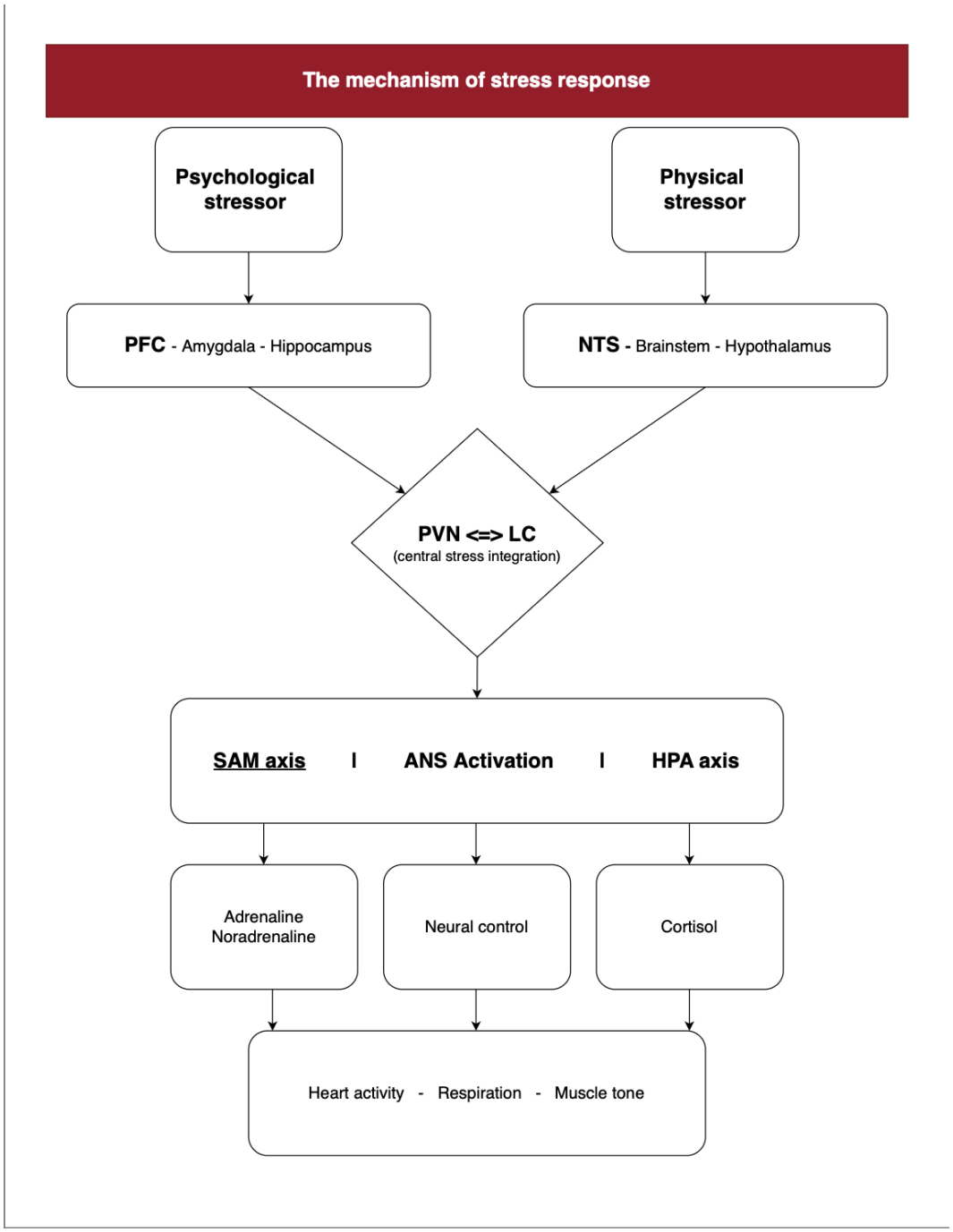

Figure 2.1: Conceptual overview of the neurophysiological stress response. Psychological and physical stressors are processed through different neural pathways but converge at central integration centers (PVN and LC), which activate the SAM axis, HPA axis, and autonomic nervous system. The resulting changes in heart activity, respiration, and muscle tone form the physiological basis for stress biomarkers. Inspired by Godoy et al. (2018).

2.4 Related Work

In recent years, significant research has been devoted to detecting and quantifying human stress using physiological signals. These studies aim to improve measurement accuracy, user comfort, and real-world applicability of stress detection systems. This

section summarizes recent developments and key challenges in the field, with a particular emphasis on multimodal sensing and radar-based approaches.

2.5 Challenges in Wearable and Physiological Stress Monitoring

Traditional stress detection methods often rely on wearable sensors that record physiological signals such as heart rate (HR), respiration rate (RR), electrodermal activity (EDA), and brain signals (EEG). While these approaches have produced promising results, they pose several challenges. First, wearing multiple sensors can negatively affect user comfort by restricting movement, irritating the skin, or causing privacy concerns. Second, many existing systems fail to effectively integrate information from multiple signal modalities. Most approaches treat physiological parameters independently rather than exploiting their interrelationships, which limits their performance and generalizability.

2.6 Radar-Based Stress Detection Approaches

Recent research has explored the use of non-contact radar systems to address these challenges. Radar-based sensing enables continuous monitoring of physiological signals without physical contact, thereby improving user experience and privacy. One promising approach is the use of ultra-wideband (UWB) radar for physiological signal extraction. Radar systems can detect subtle chest and body movements related to respiration and cardiac activity by analyzing variations in reflected radio frequency (RF) signals. This contactless sensing modality allows stress estimation based on indirect measurement of HR and RR.

2.7 Deep Stress Tracing (DST)

A recent contribution in this area is the Deep Stress Tracing (DST) framework, which implements a deep learning-based extraction module to trace human stress from raw RF data [**?**]. DST uses non-contact UWB radar to extract multimodal physiological signals and demonstrates approximately 6% higher stress detection accuracy than wearable sensors. The framework includes three primary modules: the *Signal Extraction Module* (SEM), the *Multimodal Fusion Module* (MFM), and the *Stress Tracing Module* (STM). The SEM extracts HR and RR signals from raw radar data using convolutional neural networks (CNNs) and a self-attention mechanism to address signal nonlinearity and body motion artifacts. The MFM performs multimodal signal fusion through a Coarse–Fine-Grained Parallel Feature Fusion approach, enabling cross-information sharing between detailed (HR, RR) and coarse (RF) features. Finally, the STM models temporal dynamics using long short-term memory (LSTM) networks to capture temporal dependencies and predict stress states over time.

2.8 Other Contactless and Multimodal Approaches

Other radar-based methods, such as WiStress and mmStress, employ millimeter-wave (mmWave) sensing to derive heartbeats and displacement activities (e.g., scratching or walking) for stress estimation. However, these systems typically rely on uni-modal

signal analysis and therefore lack the capability to exchange information across different physiological modalities, which limits their detection accuracy. Alternative studies have used non-contact cameras to infer stress from facial expressions. Features such as eye position, mouth movement, and facial tension are processed using machine learning algorithms, including support vector machines (SVMs), to classify stress states. While effective under controlled conditions, such methods are sensitive to lighting, facial occlusion, and head movement.

2.9 Principles of RF-Based Physiological Monitoring

The underlying principle of RF-based physiological sensing is that small body movements, such as chest expansions due to heartbeat and respiration, alter the propagation path of RF signals. These minor displacements change the phase and amplitude of the reflected signals, which can be captured by UWB radar. The radar collects the reflected signals into a structured data representation known as the Channel Impulse Response (CIR) matrix. Each row in the CIR represents the radar's response to a transmitted pulse, while each column corresponds to the temporal evolution of received signals. By analyzing variations in signal phase and frequency, physiological parameters such as HR and RR can be extracted. Recent work has applied these radar-derived features to machine learning models to continuously estimate stress levels as a binary or continuous variable over time, with the aim of enabling reliable, non-invasive stress monitoring.

2.10 Summary

In summary, while wearable sensors and traditional physiological measurements have provided valuable insights into stress detection, they are often intrusive and limited in multimodal fusion capability. Radar-based systems, particularly those employing deep learning techniques such as DST, represent a promising advancement toward non-contact, multimodal stress estimation. The present project builds upon this foundation by combining radar-derived physiological features with HRV and EDA data, integrating them within a machine learning framework to enhance stress detection accuracy and reliability.

2.11 Methods

The project employs a multi-sensor setup combining the XeThru X4M200 radar sensor with wearable devices for ECG and EDA measurement. Radar signals will be processed to extract physiological features such as respiration rate and motion patterns, while HRV and EDA data provide markers of sympathetic nervous system activity. A synchronized data collection pipeline will gather multimodal data under controlled and real-world stress-inducing scenarios, with stress levels validated using self-reported measures.
Feature extraction and preprocessing will focus on identifying stress-relevant metrics from each modality, such as respiratory variability, skin conductance responses, and heart rate patterns. Sensor fusion techniques, including feature fusion and decision fusion, will combine these features. Machine learning models such as deep neural networks will be trained and validated to predict stress levels. The models will be optimized using cross-validation and evaluated for generalizability to unseen data.

2.11.1 Stress Induction Protocol

Background

Measuring and tracking stress in individuals involves recognizing that numerous factors can influence stress levels. Different individuals may experience and respond to stress in unique ways. Therefore, it is essential to establish a consistent baseline for stress induction in the form of a “stress test” that is uniform across all participants in this research.

Objective Definition

This research project aims to trace human stress using UWB radar in combination with several other physiological sensors. The extracted signals will be processed and refined using machine learning algorithms to achieve the highest possible precision. The standardized stress induction protocol, together with self-reported stress assessments, is used to ensure that the stress levels observed in participants are a result of the "stress test" rather than external factors.

Participant Inclusion Criteria

- **Age:** 18 to 50 years old
- **Gender:** Equally distributed (10 males and 10 females)
- **Health status:** Good health

Ethical Considerations

Ethical Approval. This research project has obtained ethical approval from the Biomedical group at the University of Oslo. A complete research proposal, including objectives, methodology, and risk assessments, has been submitted for review, and thus secured approval.

Confidentiality. Participant confidentiality is a priority throughout the study. The data collected will be securely stored on password-protected devices and anonymized to protect personal identifiers. Access to the data will be limited to the research team, and the findings will be presented in aggregate form to ensure that individual identities remain confidential.

Informed Consent. A detailed informed consent form will be developed, outlining the study’s purpose, procedures, potential risks, and benefits. Participants will be informed that their participation is entirely voluntary and that they have the right to withdraw from the study at any time without any adverse consequences. The consent form will be reviewed with each participant, and they will have the opportunity to ask questions before signing.

Stress Induction

In this process, we aim to trigger mental, emotional or physical strain through activities that require cognitive effort, problem-solving, and critical thinking. For this, we will use a test divided into three parts: a preparation period, a simple question-answering task (short interview), a challenging mental arithmetic task and a physical exercise.

1. Preparation Period

Objective. In this first part, the examiner will go through the testing procedure with the participant, explain how it works, and answer questions. The examiner should make sure that the stress-measuring equipment is set up and calibrated. The candidate will then be asked to relax as much as possible in order to achieve a "no stress" state.

Procedure. After completing the relaxation phase, the participant will be asked to fill out a Visual Analog Scale (VAS) for Stress on a paper, by simply rating his/her stress level on a scale from 0 to 100. When asked "How stressed are you right now?", 0 = no stress and 100 = highest level of stress. The stress levels will be measured after the relaxation time and once the VAS is filled out.

Environment Setup. The participant will be seated comfortably in a quiet and distraction-free room. Only the participant and the examiner (researcher) are allowed in the room during the stress-inducing test and the measurements.

Briefing. A short briefing will be carried out where the examiner explains the procedures and what is expected from the participant. The participant will have the opportunity to ask questions if anything is unclear.

Calibration. Physiological sensors for measuring ECG, EDA, and HRV are attached to the participant. The examiner calibrates the measurement devices and ensures that the UWB radar is working properly.

Relaxation Exercise. The participant will be asked to sit and relax for 5 minutes focusing on breathing — slow and deep breaths, inhaling through the nose, holding briefly, and exhaling through the mouth.

2. Simple Questions Answering Task (Short Interview)

Objective. This task serves as a transition from a relaxed state to a "stressful state". It aims to engage the participant verbally through straightforward questions under a mildly stressful context.

Procedure. After answering this part, the participant will be asked to fill out the VAS for Stress. The stress levels will be measured after all questions or speeches are completed and the VAS is filled out, with no rest in between.

Social-Evaluative Speech. Participants deliver a 3-minute speech on why they would be a suitable candidate for a competitive job. They are told their performance will be evaluated and (if ethically approved) video recorded. The examiner maintains a neutral, evaluative demeanor, takes notes, and does not provide encouragement.

3. Mental Arithmetic Task

Objective. This task induces mental stress in the participant through cognitive challenge. The difficulty of the exercises is sufficient to cause a reasonable stress level, while maintaining participant wellbeing.

Procedure. The participant should answer each question as fast as possible within the given time limit. After answering each question, the participant will be asked to fill out the VAS for Stress.

Exercises.

1. Calculate the equation (1 min): $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 =$
2. Calculate the equation (1 min): $1 + (2 \times 3) - 4 + 8 - (2/2) =$
3. Choose the correct alternative (1 min): $4^4 =$ (a) 16 (b) 64 (c) 244 (d) 256
4. Word problem (1 min): Ivan has 6 apples. Georgi has 2 oranges. Ivan gives 4 apples to Georgi. Then Georgi eats one orange and buys 3 more apples. Georgi gives back 5 apples to Ivan. How many apples does Georgi have? How many oranges does Ivan have?
5. : In the following boxes the background color of the equation, which is equal to 40, is? (1 min)

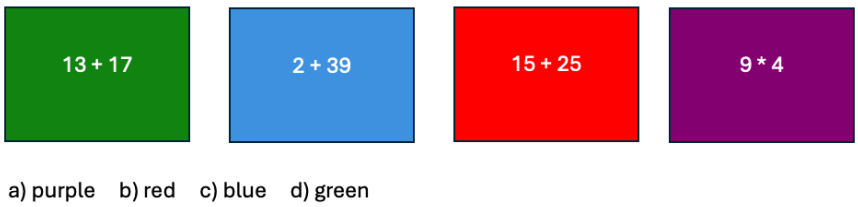

Figure 2.2: Exercise 5

6. I will ask you to subtract 7 from a starting number (e.g., 100) continuously and say each result out loud. Please work quickly and accurately. I will time you and there will be a short time limit for each response. If you make an error, I will interrupt and ask you to continue from the correct number. Do you have any questions? Ready? Start now."

Stress Measurement. The stress levels will be measured after all questions are answered and the VAS is filled out, with no rest period in between.

4. Physical Stress-Inducing Technique

Objective. In this phase, participants will engage in tasks designed to induce a stress response through physical exertion. By measuring physiological responses to these physical challenges, we aim to gather information on the differences in stress levels generated by physical versus mental stressors.

Procedure. The participant should perform every exercise as fast as possible during the given time interval. After completing all exercises, the participant will fill out the VAS for Stress. A rest of 2 minutes will be given prior to stress measurement to allow physiological recovery.

Exercises.

1. **Burpees:** Do as many burpees as possible in 20 seconds, followed by 20 seconds of rest.

2. **Coordination in 4 Directions:** React as fast as possible to the examiner's cues. Two cones are placed 2 meters apart; the participant starts in the middle. When the examiner points left or right, touch that cone and return. Up = jump, Down = lie on the stomach and return. Work for 20 seconds, rest for 20 seconds.

3. **Mountain Climbers:** Do as many repetitions as possible in 20 seconds.

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